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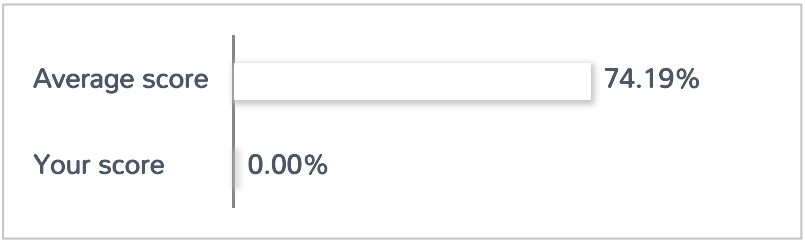
Practice Set 5

Your results are here!! for" Databricks Certified Data Engineer Associate Practice Test 5 "

0 of 50 questions answered correctly

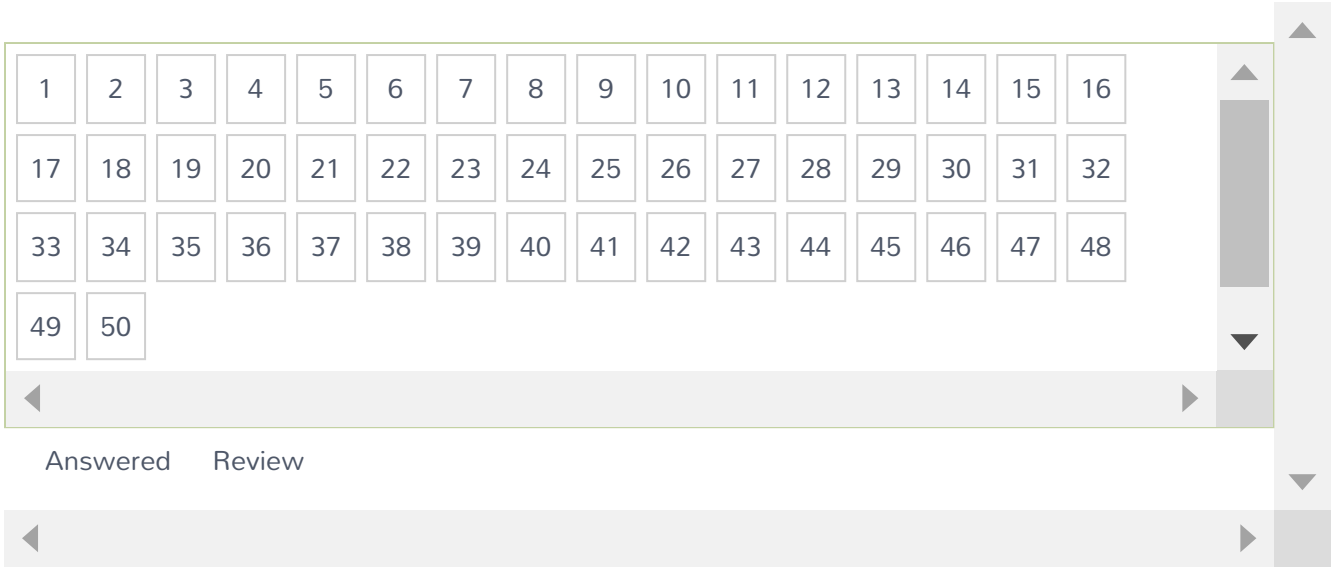
Your time: 00:00:03

Your Final Score is : 0
You have attempted : 0
Number of Correct Questions : 0 and scored 0
Number of Incorrect Questions : 0 and Negative marks 0



You can review your answers by clicking on “View Answers” option.
Important Note : Open Reference Documentation Links in New Tab (Right Click and Open in New Tab).

Restart Test View Answers



1. Question

The data engineer team has a DLT pipeline that updates all the tables at defined intervals until manually stopped. The compute resources terminate when the pipeline is stopped.

Which of the following best describes the execution modes of this DLT pipeline ?

- The DLT pipeline executes in Continuous Pipeline mode under Production mode.
- The DLT pipeline executes in Continuous Pipeline mode under Development mode.
- The DLT pipeline executes in Triggered Pipeline mode under Production mode.
- The DLT pipeline executes in Triggered Pipeline mode under Development mode.
- More information is needed to determine the correct response

Unattempted

Continuous pipelines update tables continuously as input data changes. Once an update is started, it continues to run until the pipeline is shut down.

In Production mode, the Delta Live Tables system:

Terminates the cluster immediately when the pipeline is stopped.

Restarts the cluster for recoverable errors (e.g., memory leak or stale credentials).

Retries execution in case of specific errors (e.g., a failure to start a cluster)

Reference:

https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-concepts.html

2. Question

A data engineer has defined the following data quality constraint in a Delta Live Tables pipeline:

CONSTRAINT valid_id EXPECT (id IS NOT NULL) _____

Fill in the above blank so records violating this constraint will be added to the target table, and reported in metrics

- ON VIOLATION ADD ROW
- ON VIOLATION FAIL UPDATE

ON VIOLATION SUCCESS UPDATE

ON VIOLATION NULL

There is no need to add ON VIOLATION clause. By default, records violating the constraint will be kept, and reported as invalid in the event log

Unattempted

By default, records that violate the expectation are added to the target dataset along with valid records, but violations will be reported in the event log

Reference:

https://learn.microsoft.com/en-us/azure/databricks/workflows/delta-live-tables/delta-live-tables-expectations

3. Question

The data engineer team has a DLT pipeline that updates all the tables once and then stops. The compute resources of the pipeline continue running to allow for quick testing.

Which of the following best describes the execution modes of this DLT pipeline ?

The DLT pipeline executes in Continuous Pipeline mode under Production mode.

The DLT pipeline executes in Continuous Pipeline mode under Development mode.

The DLT pipeline executes in Triggered Pipeline mode under Production mode.

The DLT pipeline executes in Triggered Pipeline mode under Development mode.

More information is needed to determine the correct response

Unattempted

Triggered pipelines update each table with whatever data is currently available and then they shut down.

In Development mode, the Delta Live Tables system ease the development process by

Reusing a cluster to avoid the overhead of restarts. The cluster runs for two hours when development mode is enabled.

Disabling pipeline retries so you can immediately detect and fix errors.

Reference:

https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-concepts.html

4. Question

Which of the following will utilize Gold tables as their source?

Silver tables

Auto loader

Bronze tables

Dashboards

Streaming jobs

Unattempted

Gold tables provide business level aggregates often used for reporting and dashboarding, or even for Machine learning

Reference:

https://www.databricks.com/glossary/medallion-architecture

5. Question

Which of the following code blocks can a data engineer use to query the existing streaming table events ?

spark.readStream("events")
spark.read
.table("events")

spark.readStream
.table("events")

spark.readStream()
.table("events")

spark.stream
.read("events")

Unattempted

Delta Lake is deeply integrated with Spark Structured Streaming. You can load tables as a stream using: spark.readStream.table(table_name)

Reference: https://docs.databricks.com/structured-streaming/delta-lake.html

6. Question

In multi-hop architecture, which of the following statements best describes the Bronze layer ?

It maintains data that powers analytics, machine learning, and production applications

It maintains raw data ingested from various sources

It represents a filtered, cleaned, and enriched version of data

It provides business-level aggregated version of data

It provides a more refined view of the data.

Unattempted

Bronze tables contain data in its rawest format ingested from various sources (e.g., JSON files, Operational Databaes, Kakfa stream, ...)

Reference:
https://www.databricks.com/glossary/medallion-architecture

7. Question

Given the following Structured Streaming query
(spark.readStream
.format(“cloudFiles“)
.option(“cloudFiles.format“, “json“)
.load(ordersLocation)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.table(“uncleanedOrders“)
)
Which of the following best describe the purpose of this query in a multi-hop architecture?

- The query is performing raw data ingestion into a Bronze table
- The query is performing a hop from a Bronze table to a Silver table
- The query is performing a hop from Silver table to a Gold table
- The query is performing data transfer from a Gold table into a production application
- This query is performing data quality controls prior to Silver layer

Unattempted
The query here is using Autoloader (cloudFiles) to load raw json data from ordersLocation into the Bronze table uncleanedOrders
References:
https://www.databricks.com/glossary/medallion-architecture
https://docs.databricks.com/ingestion/auto-loader/index.html

8. Question

A data engineer has the following query in a Delta Live Tables pipeline:
CREATE LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM cleaned_sales
GROUP BY store_id
The pipeline is failing to start due to an error in this query
Which of the following changes should be made to this query to successfully start the DLT pipeline ?

- CREATE STREAMING TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM LIVE.cleaned_sales
GROUP BY store_id

CREATE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM LIVE.cleaned_sales
GROUP BY store_id

CREATE LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM LIVE.cleaned_sales
GROUP BY store_id

CREATE STREAMING LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM cleaned_sales
GROUP BY store_id

CREATE STREAMING LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM STREAM(cleaned_sales)
GROUP BY store_id

Unattempted

In DLT pipelines, we use the CREATE LIVE TABLE syntax to create a table with SQL. To query another live table, prepend the LIVE. keyword to the table name.

CREATE LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM LIVE.cleaned_sales
GROUP BY store_id

Reference: https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-sql-ref.html

9. Question

A data engineer has defined the following data quality constraint in a Delta Live Tables pipeline:

CONSTRAINT valid_id EXPECT (id IS NOT NULL) _____

Fill in the above blank so records violating this constraint will be dropped, and reported in metrics

ON VIOLATION DROP ROW

ON VIOLATION FAIL UPDATE

ON VIOLATION DELETE ROW

ON VIOLATION DISCARD ROW

There is no need to add ON VIOLATION clause. By default, records violating the constraint will be discarded, and reported as invalid in the event log

Unattempted

With ON VIOLATION DROP ROW, records that violate the expectation are dropped, and violations are reported in the event log

Reference:
https://learn.microsoft.com/en-us/azure/databricks/workflows/delta-live-tables/delta-live-tables-expectations

10. Question

Which of the following compute resources is available in Databricks SQL ?

- Single-node clusters
- Multi-nodes clusters
- On-premises clusters
- **SQL warehouses**
- SQL engines

Unattempted

Compute resources are infrastructure resources that provide processing capabilities in the cloud. A SQL warehouse is a compute resource that lets you run SQL commands on data objects within Databricks SQL.

Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html

11. Question

Which of the following is the benefit of using the Auto Stop feature of Databricks SQL warehouses ?

- Improves the performance of the warehouse by automatically stopping ideal services
- **Minimizes the total running time of the warehouse**
- Provides higher security by automatically stopping unused ports of the warehouse
- Increases the availability of the warehouse by automatically stopping long-running SQL queries
- Databricks SQL does not have Auto Stop feature

Unattempted

The Auto Stop feature stops the warehouse if it's idle for a specified number of minutes.

Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html

12. Question

Which of the following alert destinations is Not supported in Databricks SQL ?

- Slack
- Webhook
- SMS
- Microsoft Teams
- Email

Unattempted

SMS is not supported as an alert destination in Databricks SQL . While, email, webhook, Slack, and Microsoft Teams are supported alert destinations in Databricks SQL.
Reference: https://docs.databricks.com/sql/admin/alert-destinations.html

13. Question

A data engineer wants to increase the cluster size of an existing Databricks SQL warehouse.
Which of the following is the benefit of increasing the cluster size of Databricks SQL warehouses ?

- Improves the latency of the queries execution
- Speeds up the start up time of the SQL warehouse
- Reduces cost since large clusters use Spot instances
- The cluster size of SQL warehouses is not configurable. Instead, they can increase the number of clusters
- The cluster size can not be changed for existing SQL warehouses. Instead, they can enable the auto-scaling option.

Unattempted

Cluster Size represents the number of cluster workers and size of compute resources available to run your queries and dashboards. To reduce query latency, you can increase the cluster size.
Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html#cluster-size-1

14. Question

Which of the following statements best describes Auto Loader ?

- Auto loader allows applying Change Data Capture (CDC) feed to update tables based on changes captured in source data.
- Auto loader monitors a source location, in which files accumulate, to identify and ingest only new arriving files with each command run. While the files that have already been ingested in previous runs are skipped.

- Auto loader allows cloning a source Delta table to a target destination at a specific version.
- Auto loader defines data quality expectations on the contents of a dataset, and reports the records that violate these expectations in metrics.
- Auto loader enables efficient insert, update, deletes, and rollback capabilities by adding a storage layer that provides better data reliability to data lakes.

Unattempted

Auto Loader incrementally and efficiently processes new data files as they arrive in cloud storage.
Reference: https://docs.databricks.com/ingestion/auto-loader/index.html

15. Question

Which part of the Databricks Platform can a data engineer use to grant permissions on tables to users ?

- Data Studio
- Cluster event log
- Workflows
- DBFS
- **Data Explorer**

Unattempted

Data Explorer in Databricks SQL allows you to manage data object permissions. This includes granting privileges on tables and databases to users or groups of users.
Reference: https://docs.databricks.com/security/access-control/data-acl.html#data-explorer

16. Question

Which of the following commands can a data engineer use to grant full permissions to the HR team on the table employees ?

- GRANT FULL PRIVILEGES ON TABLE employees TO hr_team
- GRANT FULL PRIVILEGES ON TABLE hr_team TO employees
- **GRANT ALL PRIVILEGES ON TABLE employees TO hr_team**
- GRANT ALL PRIVILEGES ON TABLE hr_team TO employees
- GRANT SELECT, MODIFY, CREATE, READ_METADATA ON TABLE employees TO hr_team

Unattempted

ALL PRIVILEGES is used to grant full permissions on an object to a user or group of users. It is translated into all the below privileges:
SELECT
CREATE

MODIFY
USAGE
READ_METADATA
Reference: https://docs.databricks.com/security/access-control/table-acls/object-privileges.html#privileges

17. Question

A data engineer uses the following SQL query:
GRANT MODIFY ON TABLE employees TO hr_team
Which of the following describes the ability given by the MODIFY privilege ?

- It gives the ability to add data from the table
- It gives the ability to delete data from the table
- It gives the ability to modify data in the table
- **All the above abilities are given by the MODIFY privilege**
- None of these options correctly describe the ability given by the MODIFY privilege

Unattempted
The MODIFY privilege gives the ability to add, delete, and modify data to or from an object.
Reference: https://docs.databricks.com/security/access-control/table-acls/object-privileges.html#privileges

18. Question

One of the foundational technologies provided by the Databricks Lakehouse Platform is an open-source, file-based storage format that brings reliability to data lakes.
Which of the following technologies is being described in the above statement?

- Delta Lives Tables (DLT)
- **Delta Lake**
- Apache Spark
- Unity Catalog
- Photon

Unattempted
Delta Lake is an open source technology that extends Parquet data files with a file-based transaction log for ACID transactions that brings reliability to data lakes.
Reference: https://docs.databricks.com/delta/index.html

19. Question

Which of the following commands can a data engineer use to purge stale data files of a Delta table?

- DELETE
- GARBAGE COLLECTION
- CLEAN
- **VACUUM**
- OPTIMIZE

Unattempted

The VACUUM command deletes the unused data files older than a specified data retention period.

Reference: https://docs.databricks.com/sql/language-manual/delta-vacuum.html

20. Question

In Databricks Repos, which of the following operations a data engineer can use to save local changes of a repo to its remote repository ?

- Create Pull Request
- Commit & Pull
- **Commit & Push**
- Merge & Push
- Merge & Pull

Unattempted

Commit & Push is used to save the changes on a local repo, then uploads this local repo content to the remote repository.

References:

https://docs.databricks.com/repos/index.html

https://github.com/git-guides/git-push

21. Question

In Delta Lake tables, which of the following is the primary format for the transaction log files?

- Delta
- Parquet
- **JSON**
- Hive-specific format
- XML

Unattempted

Delta Lake builds upon standard data formats. Delta lake table gets stored on the storage in one or more data files in Parquet format, along with transaction logs in JSON format.
Reference: https://docs.databricks.com/delta/index.html

22. Question

Which of the following functionalities can be performed in Databricks Repos ?

- Create pull requests
- Create new remote Git repositories
- Delete branches
- Create CI/CD pipelines
- **Pull from a remote Git repository**

Unattempted

Databricks Repos supports git Pull operation. It is used to fetch and download content from a remote repository and immediately update the local repo to match that content.
References:
https://docs.databricks.com/repos/index.html
https://github.com/git-guides/git-pull

23. Question

Which of the following tasks is not supported by Databricks Repos, and must be performed in your Git provider ?

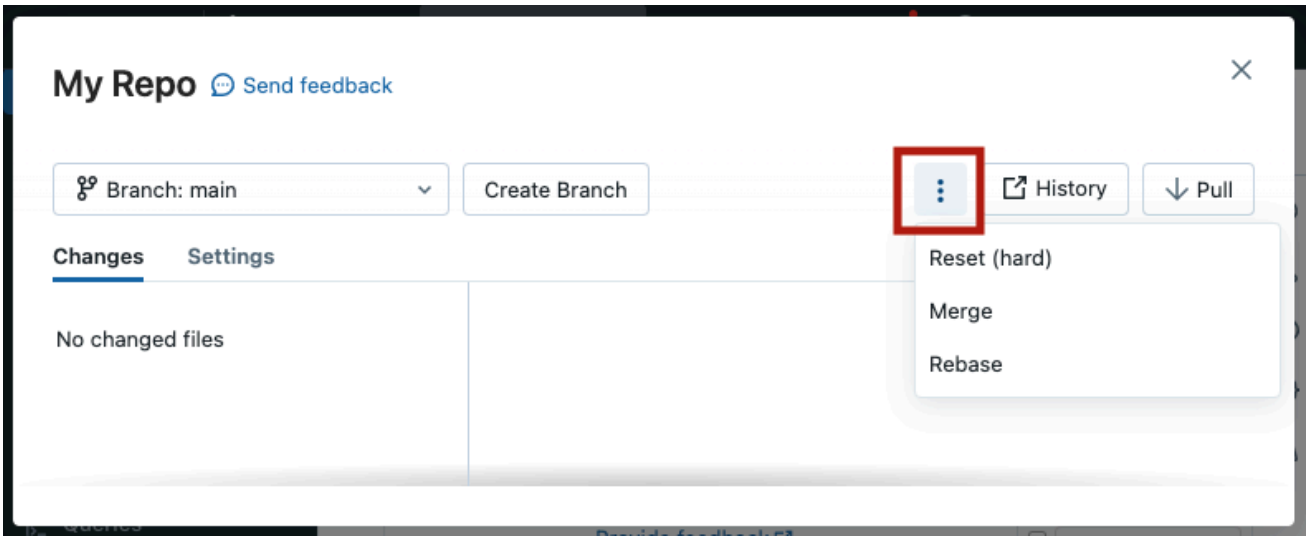
- Clone, push to, or pull from a remote Git repository.
- Create and manage branches for development work.
- Create notebooks, and edit notebooks and other files.
- Visually compare differences upon commit.
- **Delete branches**

Unattempted

The following tasks are not supported by Databricks Repos, and must be performed in your Git provider:

- Create a pull request
- Delete branches
- Merge and rebase branches *

* NOTE: Recently, merge and rebase branches have become supported in Databricks Repos. However, this may still not be updated in the current exam version.



Reference: https://docs.databricks.com/repos/index.html

24. Question

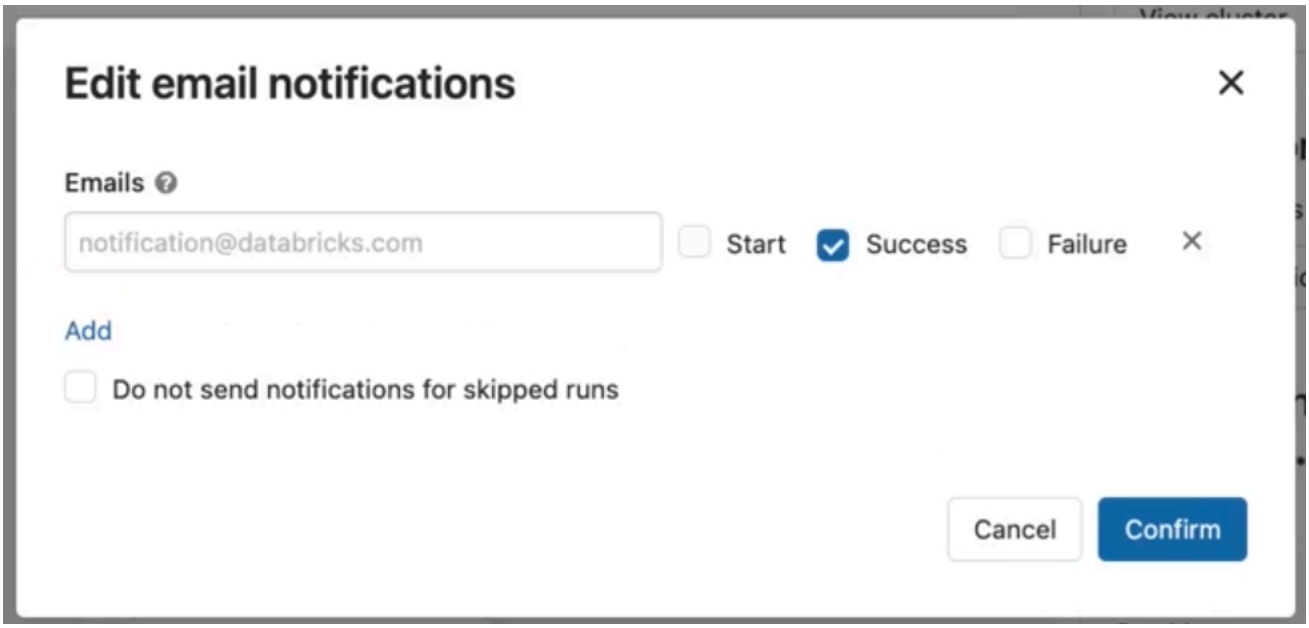
A data engineering team has a long-running multi-tasks Job. The team members need to be notified when the run of this job completes.

Which of the following approaches can be used to send emails to the team members when the job completes ?

- They can use Job API to programmatically send emails according to each task status
- **They can configure email notifications settings in the job page**
- There is no way to notify users when the job completes
- Only Job owner can be configured to be notified when the job completes
- They can configure email notifications settings per notebook in the task page

Unattempted

Databricks Jobs supports email notifications to be notified in the case of job start, success, or failure. Simply, click **Edit email notifications** from the details panel in the Job page. From there, you can add one or more email addresses.



Reference: https://docs.databricks.com/workflows/jobs/jobs.html#alerts-job

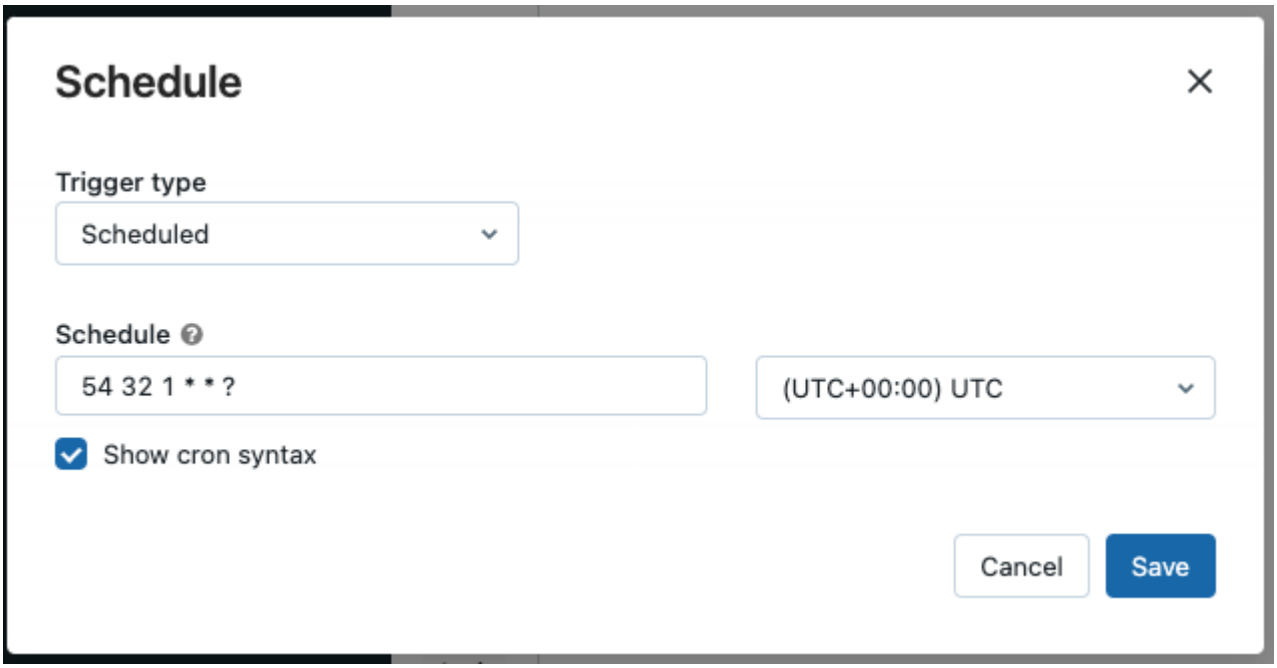
25. Question

Which of the following describes Cron syntax in Databricks Jobs ?

- It's an expression to represent the maximum concurrent runs of a job
- It's an expression to represent complex job schedule that can be defined programmatically
- It's an expression to represent the retry policy of a job
- It's an expression to describe the email notification events (start, success, failure)
- It's an expression to represent the run timeout of a job

Unattempted

To define a schedule for a Databricks job, you can either interactively specify the period and starting time, or write a Cron Syntax expression. The Cron Syntax allows to represent complex job schedule that can be defined programmatically



Reference: https://docs.databricks.com/workflows/jobs/jobs.html#schedule-a-job

26. Question

Fill in the below blank to successfully create a table in Databricks using data from an existing PostgreSQL database:

CREATE TABLE employees
USING _____
OPTIONS (
url “jdbc:postgresql:dbserver“,
dbtable “employees“
)

- org.apache.spark.sql.jdbc
- postgresql
- DELTA
- dbserver
- cloudfiles

Unattempted

Using the JDBC library, Spark SQL can extract data from any existing relational database that supports JDBC. Examples include mysql, postgres, SQLite, and more.
Reference: https://learn.microsoft.com/en-us/azure/databricks/external-data/jdbc

27. Question

A data engineer is trying to use Delta time travel to rollback a table to a previous version, but the data engineer received an error that the data files are no longer present.
Which of the following commands was run on the table that caused deleting the data files?

- VACUUM
- OPTIMIZE
- ZORDER BY
- DEEP CLONE
- DELETE

Unattempted

Running the VACUUM command on a Delta table deletes the unused data files older than a specified data retention period. As a result, you lose the ability to time travel back to any version older than that retention threshold.

Reference: https://docs.databricks.com/sql/language-manual/delta-vacuum.html

28. Question

In Delta Lake tables, which of the following is the primary format for the data files?

- Delta
- Parquet
- JSON
- Hive-specific format
- Both, Parquet and JSON

Unattempted

Delta Lake builds upon standard data formats. Delta lake table gets stored on the storage in one or more data files in Parquet format, along with transaction logs in JSON format.

Reference: https://docs.databricks.com/delta/index.html

29. Question

Which of the following locations hosts the Databricks web application ?

- Data plane
- Control plane
- Databricks Filesystem
- Databricks-managed cluster
- Customer Cloud Account

Unattempted

According to the Databricks Lakehouse architecture, Databricks workspace is deployed in the control plane along with Databricks services like Databricks web application (UI), Cluster manager, workflow

service, and notebooks.
Reference: https://docs.databricks.com/getting-started/overview.html

30. Question

In Databricks Repos, which of the following operations a data engineer can use to update the local version of a repo from its remote Git repository ?

- Clone
- Commit
- Merge
- Push
- **Pull**

Unattempted
The git Pull operation is used to fetch and download content from a remote repository and immediately update the local repository to match that content.
References:
https://docs.databricks.com/repos/index.html
https://github.com/git-guides/git-pull

31. Question

According to the Databricks Lakehouse architecture, which of the following is located in the customer's cloud account?

- Databricks web application
- Notebooks
- Repos
- **Cluster virtual machines**
- Workflows

Unattempted
When the customer sets up a Spark cluster, the cluster virtual machines are deployed in the data plane in the customer's cloud account.
Reference: https://docs.databricks.com/getting-started/overview.html

32. Question

Which of the following best describes Databricks Lakehouse?

- **Single, flexible, high-performance system that supports data, analytics, and machine learning**

workloads.

- Reliable data management system with transactional guarantees for organization's structured data.
- Platform that helps reduce the costs of storing organization's open-format data files in the cloud.
- Platform for developing increasingly complex machine learning workloads using a simple, SQL-based solution.
- Platform that scales data lake workloads for organizations without investing on-premises hardware.

Unattempted

Databricks Lakehouse is a unified analytics platform that combines the best elements of data lakes and data warehouses. So, in the Lakehouse, you can work on data engineering, analytics, and AI, all in one platform.

Reference: https://www.databricks.com/glossary/data-lakehouse

33. Question

If the default notebook language is SQL, which of the following options a data engineer can use to run a Python code in this SQL Notebook ?

- They need first to import the python module in a cell
- This is not possible! They need to change the default language of the notebook to Python
- Databricks detects cells language automatically, so they can write Python syntax in any cell
- They can add %language magic command at the start of a cell to force language detection.
- **They can add %python at the start of a cell.**

Unattempted

By default, cells use the default language of the notebook. You can override the default language in a cell by using the language magic command at the beginning of a cell. The supported magic commands are: %python, %sql, %scala, and %r.

Reference: https://docs.databricks.com/notebooks/notebooks-code.html

34. Question

Which of the following statements is Not true about Delta Lake ?

- Delta Lake provides ACID transaction guarantees
- Delta Lake provides scalable data and metadata handling
- Delta Lake provides audit history and time travel
- **Delta Lake builds upon standard data formats: Parquet + XML**
- Delta Lake supports unified streaming and batch data processing

Unattempted

It is not true that Delta Lake builds upon XML format. It builds upon Parquet and JSON formats

Reference: https://docs.databricks.com/delta/index.html

35. Question

How long is the default retention period of the VACUUM command ?

- 0 days
- **7 days**
- 30 days
- 90 days
- 365 days

Unattempted

By default, the retention threshold of the VACUUM command is 7 days. This means that VACUUM operation will prevent you from deleting files less than 7 days old, just to ensure that no long-running operations are still referencing any of the files to be deleted.

Reference: https://docs.databricks.com/sql/language-manual/delta-vacuum.html

36. Question

The data engineering team has a Delta table called employees that contains the employees personal information including their gross salaries.

Which of the following code blocks will keep in the table only the employees having a salary greater than 3000 ?

- DELETE FROM employees WHERE salary > 3000;
- SELECT CASE WHEN salary <= 3000 THEN DELETE ELSE UPDATE END FROM employees;
- UPDATE employees WHERE salary > 3000 WHEN MATCHED SELECT;
- UPDATE employees WHERE salary <= 3000 WHEN MATCHED DELETE;
- **DELETE FROM employees WHERE salary <= 3000;**

Unattempted

In order to keep only the employees having a salary greater than 3000, we must delete the employees having salary less than or equal 3000. To do so, use the DELETE statement:

DELETE FROM table_name WHERE condition;

Reference: https://docs.databricks.com/sql/language-manual/delta-delete-from.html

37. Question

A data engineer wants to create a relational object by pulling data from two tables. The relational object must be used by other data engineers in other sessions on the same cluster only. In order to save on storage costs, the date engineer wants to avoid copying and storing physical data.
Which of the following relational objects should the data engineer create?

- Temporary view
- External table
- Managed table
- Global Temporary view
- View

Unattempted
In order to avoid copying and storing physical data, the data engineer must create a view object. A view in databricks is a virtual table that has no physical data. It's just a saved SQL query against actual tables.
The view type should be Global Temporary view that can be accessed in other sessions on the same cluster. Global Temporary views are tied to a cluster temporary database called global_temp.
Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-view.html

38. Question

A data engineer has developed a code block to completely reprocess data based on the following if-condition in Python:
if process_mode = "init" and not is_table_exist:
print("Start processing ...")
This if-condition is returning an invalid syntax error.
Which of the following changes should be made to the code block to fix this error ?

- if process_mode = "init" & not is_table_exist:
print("Start processing ...")
- if process_mode = "init" and not is_table_exist = True:
print("Start processing ...")
- if process_mode = "init" and is_table_exist = False:
print("Start processing ...")
- if (process_mode = "init") and (not is_table_exist):
print("Start processing ...")
- if process_mode == "init" and not is_table_exist:
print("Start processing ...")

Unattempted
Python if statement looks like this in its simplest form:
if :

Python supports the usual logical conditions from mathematics:
Equals: a == b
Not Equals: a != b
<, <=, >, >=
To combine conditional statements, you can use the following logical operators:
and
or
The negation operator in Python is: not
Reference: https://www.w3schools.com/python/python_conditions.asp

39. Question

Which of the following commands can a data engineer use to compact small data files of a Delta table into larger ones ?

- PARTITION BY
- ZORDER BY
- COMPACT
- VACUUM
- **OPTIMIZE**

Unattempted
Delta Lake can improve the speed of read queries from a table. One way to improve this speed is by compacting small files into larger ones. You trigger compaction by running the OPTIMIZE command
Reference: https://docs.databricks.com/sql/language-manual/delta-optimize.html

40. Question

Which of the following commands can a data engineer use to create a new table along with a comment ?

- **CREATE TABLE payments**
COMMENT “This table contains sensitive information“
AS SELECT * FROM bank_transactions
- CREATE TABLE payments
COMMENT(“This table contains sensitive information“)
AS SELECT * FROM bank_transactions
- CREATE TABLE payments
AS SELECT * FROM bank_transactions
COMMENT “This table contains sensitive information“
- CREATE TABLE payments
AS SELECT * FROM bank_transactions
COMMENT(“This table contains sensitive information“)

COMMENT(“This table contains sensitive information“)
CREATE TABLE payments
AS SELECT * FROM bank_transactions

Unattempted
The CREATE TABLE clause supports adding a descriptive comment for the table. This allows for easier discovery of table contents.
Syntax:
CREATE TABLE table_name
COMMENT “here is a comment“
AS query
Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-table-using.html

41. Question

A junior data engineer usually uses INSERT INTO command to write data into a Delta table. A senior data engineer suggested using another command that avoids writing of duplicate records.
Which of the following commands is the one suggested by the senior data engineer ?

MERGE INTO
APPLY CHANGES INTO
UPDATE
COPY INTO
INSERT OR OVERWRITE

Unattempted
MERGE INTO allows to merge a set of updates, insertions, and deletions based on a source table into a target Delta table. With MERGE INTO, you can avoid inserting the duplicate records when writing into Delta tables.
References:
https://docs.databricks.com/sql/language-manual/delta-merge-into.html
https://docs.databricks.com/delta/merge.html#data-deduplication-when-writing-into-delta-tables

42. Question

A data engineer is designing a Delta Live Tables pipeline. The source system generates files containing changes captured in the source data. Each change event has metadata indicating whether the specified record was inserted, updated, or deleted. In addition to a timestamp column indicating the order in which the changes happened. The data engineer needs to update a target table based on these change events.
Which of the following commands can the data engineer use to best solve this problem?

MERGE INTO
APPLY CHANGES INTO

- UPDATE
- COPY INTO
- cloud_files

Unattempted

The events described in the question represent Change Data Capture (CDC) feed. CDC is logged at the source as events that contain both the data of the records along with metadata information:
1. Operation column indicating whether the specified record was inserted, updated, or deleted
2. Sequence column that is usually a timestamp indicating the order in which the changes happened
You can use the APPLY CHANGES INTO statement to use Delta Live Tables CDC functionality
Reference: https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-cdc.html

43. Question

In PySpark, which of the following commands can you use to query the Delta table employees created in Spark SQL?

- pyspark.sql.read(SELECT * FROM employees)
- spark.sql("employees")
- spark.format("sql").read("employees")
- **spark.table("employees")**
- Spark SQL tables can not be accessed from PySpark

Unattempted

spark.table() function returns the specified Spark SQL table as a PySpark DataFrame
Reference:
https://spark.apache.org/docs/2.4.0/api/python/_modules/pyspark/sql/session.html#SparkSession.table

44. Question

Which of the following code blocks can a data engineer use to create a user defined function (UDF) ?

- CREATE FUNCTION plus_one(value INTEGER)
RETURN value +1
CREATE UDF plus_one(value INTEGER)
RETURNS INTEGER
RETURN value +1;
- CREATE UDF plus_one(value INTEGER)
RETURN value +1;
- **CREATE FUNCTION plus_one(value INTEGER)**
RETURNS INTEGER

RETURN value +1;

CREATE FUNCTION plus_one(value INTEGER)
RETURNS INTEGER
value +1;

Unattempted
The correct syntax to create a UDF is:
CREATE [OR REPLACE] FUNCTION function_name ([parameter_name data_type [, ...]])
RETURNS data_type
RETURN { expression | query }
Reference: https://docs.databricks.com/udf/index.html

45. Question

When dropping a Delta table, which of the following explains why only the table's metadata will be deleted, while the data files will be kept in the storage ?

The table is deep cloned

The table is external

The user running the command has no permission to delete the data files

The table is managed

Delta prevents deleting files less than retention threshold, just to ensure that no long-running operations are still referencing any of the files to be deleted

Unattempted
External (unmanaged) tables are tables whose data is stored in an external storage path by using a LOCATION clause.
When you run DROP TABLE on an external table, only the table's metadata is deleted, while the underlying data files are kept.
Reference: https://docs.databricks.com/lakehouse/data-objects.html#what-is-an-unmanaged-table

46. Question

Given the two tables students_course_1 and students_course_2. Which of the following commands can a data engineer use to get all the students from the above two tables without duplicate records ?

SELECT * FROM students_course_1
CROSS JOIN
SELECT * FROM students_course_2

SELECT * FROM students_course_1
UNION

SELECT * FROM students_course_2

- SELECT * FROM students_course_1
INTERSECT
SELECT * FROM students_course_2
- SELECT * FROM students_course_1
OUTER JOIN
SELECT * FROM students_course_2
- SELECT * FROM students_course_1
INNER JOIN
SELECT * FROM students_course_2

Unattempted

With UNION, you can return the result of subquery1 plus the rows of subquery2

Syntax:

subquery1

UNION [ALL | DISTINCT]

subquery2

If ALL is specified duplicate rows are preserved.

If DISTINCT is specified the result does not contain any duplicate rows. This is the default.

Note that both subqueries must have the same number of columns and share a least common type for each respective column.

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-qry-select-setops.html

47. Question

Given the following command:

CREATE DATABASE IF NOT EXISTS hr_db ;

In which of the following locations will the hr_db database be located?

- dbfs:/user/hive/warehouse
- dbfs:/user/hive/db_hr
- dbfs:/user/hive/databases/db_hr.db
- dbfs:/user/hive/databases
- dbfs:/user/hive

Unattempted

Since we are creating the database here without specifying a LOCATION clause, the database will be created in the default warehouse directory under dbfs:/user/hive/warehouse

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-schema.html

48. Question

Fill in the below blank to get the students enrolled in less than 3 courses from array column students

SELECT
faculty_id,
students,
_____ AS few_courses_students

- FROM faculties
- TRANSFORM (students, total_courses < 3)
- TRANSFORM (students, i -> i.total_courses < 3)
- FILTER (students, total_courses < 3)
- **FILTER (students, i -> i.total_courses < 3)**
- CASE WHEN students.total_courses < 3 THEN students

Unattempted

filter(input_array, lamda_function) is a higher order function that returns an output array from an input array by extracting elements for which the predicate of a lambda function holds.

Example:

Extracting odd numbers from an input array of integers:

SELECT filter(array(1, 2, 3, 4), i -> i % 2 == 1);

output: [1, 3]

References:

https://docs.databricks.com/sql/language-manual/functions/filter.html

https://docs.databricks.com/optimizations/higher-order-lambda-functions.html

49. Question

Given the following Structured Streaming query:

(spark.table(“orders“)
.withColumn(“total_after_tax“, col(“total“)+col(“tax“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
._____
.table(“new_orders“)
)

Fill in the blank to make the query executes a micro-batch to process data every 2 minutes

- trigger(once=”2 minutes”)
- **trigger(processingTime=”2 minutes“)**
- processingTime(”2 minutes“)
- trigger(”2 minutes“)

trigger()

Unattempted

In Spark Structured Streaming, in order to process data in micro-batches at the user-specified intervals, you can use processingTime keyword. It allows to specify a time duration as a string.

Reference: https://docs.databricks.com/structured-streaming/triggers.html#configure-structured-streaming-trigger-intervals

50. Question

Which of the following is used by Auto Loader to load data incrementally?

DEEP CLONE

Multi-hop architecture

COPY INTO

Spark Structured Streaming

Databricks SQL

Unattempted

Auto Loader is based on Spark Structured Streaming. It provides a Structured Streaming source called cloudFiles.

Reference: https://docs.databricks.com/ingestion/auto-loader/index.html

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